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ABSTRACT

Depression is a psychological disorder characterized by the continuous occurrence of a low mood state. It significantly affects individuals across multiple age groups worldwide and has become a serious global health concern. Early diagnosis plays a vital role in preventing its long-term impact and can potentially save lives. Electroencephalogram (EEG) signals serve as a reliable diagnostic tool, as the analysis of brain wave patterns can reflect the underlying mental state of an individual.

A fully automated depression detection system offers advantages over manual EEG analysis, which is time-consuming, labor-intensive, and requires significant expertise. A novel computer-aided hybrid neural network model, known as DepHNN (Depression Hybrid Neural Network), has been developed for depression screening using EEG signals. The model integrates a Convolutional Neural Network (CNN) for spatial and temporal feature extraction, along with Long Short-Term Memory (LSTM) layers for learning sequential patterns. The use of a windowing technique reduces computation complexity and training time. EEG data was collected from 21 drug-free symptomatic depressed patients and 24 normal individuals using a neuroscan system. The model achieved a high classification accuracy of 99.10% and a mean absolute error (MAE) of 0.2040, demonstrating its effectiveness, low complexity, and strong potential for real-time depression detection.

CONTENTS

Acknowledgement	i
Abstract	ii
Contents	iii-iv
List of Figures	v
List of Tables	vi

CHAPTER NO	CHAPTER TITLE	PAGE NO
1	INTRODUCTION	1-8
1.1	Overview	1-3
1.2	Problem Statement	3-4
1.3	Problem Definition	5
1.4	Purpose	6
1.5	Scope	6-7
1.6	Objectives	7-8
1.7	Organization of Chapters	8
2	LITERATURE SURVEY	9-17
3	SYSTEM ANALYSIS	18-20
3.1	Existing System	18-19
3.2	Proposed System	19-20
4	SYSTEM REQUIREMENTS SPECIFICATION	21-23
4.1	Hardware Specifications	21-22
4.2	Software Specifications	22-23
5	SYSTEM DESIGN	24-35
5.1	Introduction	24-25
5.2	Feasibility Study	25-27
5.3	Architectures	28-31

5.4	Object Oriented Design	31-35
6	IMPLEMENTATION	36-47
6.1	Methodology	36
6.2	Algorithms	37-39
6.3	Modules	40-44
6.4	Code	44-47
7	TESTING	48-60
7.1	Testing	48
7.2	Types of Testing	48-56
7.2.1	Unit Testing	48-49
7.2.2	Integrated Testing	49-50
7.2.3	Functional Testing	50-51
7.2.4	System Testing	51
7.2.5	Acceptance Testing	52
7.2.6	Data Flow Testing	52-53
7.2.7	Control Flow Testing	53-54
7.2.8	White Box Testing	54
7.2.9	Black Box Testing	54-55
7.3	Test Cases	56-60
8	CONCLUSION AND FUTURE ENHANCEMENT	61-63
8.1	Conclusion	61-62
8.2	Future Enhancement	62-63
BIBLIOGRAPHY		64-65
APPENDIX A SNAPSHOTS		66-69

LIST OF FIGURES

Figure No	Name of the Figure	Page No
5.1	System Architecture	28
5.2	Model Architecture	29
5.3	LSTM Architecture	30
5.1	Flow Chart	31
5.2	Use Case Diagram	32
5.3	Activity Diagram	33
5.4	Sequence Diagram	34
9.1	Terminal Output Showing Successful Integration of TensorFlow Model in Django	66
9.2	Homepage of DepHNN Analyzer Web Application	66
9.3	About Us Page of DepHNN Analyzer Web Application	67
9.4	Contact Us Page of DepHNN Analyzer Web Application	67
9.5	EEG Signal Upload and Analysis Interface of DepHNN Analyzer	68
9.6	EEG Data File Selection Interface	68
9.7	EEG Signal Classification Result for Depression Detection	69
9.8	Confusion Matrix on Test Data.	69
9.9	Comparison of Training and Validation Accuracy during model training.	70
9.10	Training vs. Validation Loss curves indicating model convergence	70
9.11	Logout Confirmation Dialog Box	71

LIST OF TABLES

Table No	Name	Page No
7.1	Unit test case - user model	56
7.2	Unit test case - OTP generation	56
7.3	Unit test case - Django forms	56-57
7.4	Integration test case - user registration flow	57
7.5	Integration test case - prediction flow	57
7.6	Functional test case - file upload & analysis	58
7.7	System test case - end-to-end prediction	58
7.8	White box test case - control flow	58-59
7.9	Black box test case - UI testing	59
7.10	Acceptance test case - prediction accuracy	59-60

CHAPTER 1

INTRODUCTION